BROWN REPRESENTABILITY PRODUCES A RIGHT ADJOINT OF AN EXACT FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the functor to be represented is displayed and shown to satisfy the hypothesis of the representability lemma inside the span, and the two closing citations record standard consequences of that representation.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Brown representability for the derived category. Same chapter, Lemma lemma-brown. Supplies the representing object for the cohomological functor considered in the proof, which is the first half of the argument.
- Existence of an adjoint from representability. Categories, Lemma categories-lemma-adjoint-exists. Supplies that representing the functors of the form Hom(F(-), Y) by objects of the source yields a right adjoint.
- Exactness of an adjoint. Derived Categories, Lemma derived-lemma-adjoint-is-exact. Supplies that the adjoint obtained is exact, which is the second assertion.

Source reference. Original lemma 0F5X: sources/stacks-injectives/pinned-original-source.tex, source lines 2975–2975. Navigation pointer invoked at proof line 3036; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0A8B: sources/stacks-categories/pinned-original-source.tex, source lines 3603–3603. Navigation pointer invoked at proof line 3039; the counted span contains the pointer only and the cited result is not reproduced

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Source reference. Original lemma 0A8D: sources/stacks-derived/pinned-original-source.tex, source lines 2507–2507. Navigation pointer invoked at proof line 3041; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let $\mathcal{A}$ be a Grothendieck abelian category. Let $\mathcal{D}$ be a triangulated category. Let $F : D(\mathcal{A}) \to \mathcal{D}$ be an exact functor of triangulated categories which transforms direct sums into direct sums. Then F has an exact right adjoint.*

Proof. For an object Y of $\mathcal{D}$ consider the contravariant functor

$$D(\mathcal{A}) \to Ab, \quad W \mapsto \mathrm{Hom}_{\mathcal{D}}(F(W), Y)$$

This is a cohomological functor as F is exact and transforms direct sums into products as F transforms direct sums into direct sums. Thus by Lemma 0F5X we find an object X of $D(\mathcal{A})$ such that $\mathrm{Hom}_{D(\mathcal{A})}(W, X) = \mathrm{Hom}_{\mathcal{D}}(F(W), Y)$. The existence of the adjoint follows from Categories, Lemma 0A8B. Exactness follows from Derived Categories, Lemma 0A8D. □